

Alper Duran Dincer

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EDUCATION

Istanbul Technical University (ITU)

Bachelor of Science in Physics Engineering

Istanbul, Turkey

Sep. 2022 – Present

- **Coursework:** Physics in Machine Learning, Statistical Mechanics, Stochastic Processes, Numerical Techniques.

EXPERIENCE

Quantitative Research Intern

June 2026 – Sep. 2026

Ata Portföy (Asset Management)

Istanbul, Turkey

- Sole researcher on a pipeline over ~6 TB of full-depth (ITCH) **order-book data** (98 BIST equities, 100+ days); built the **data-validation and normalization layer** (schema regime changes, symbol renames, partial sessions, corrupted records) and verified the tick-size proxy against the exchange's step table (120/120).
- Designed and ran a **pre-registered 7-test validation battery** (placebo, permutation, decile monotonicity, hourly and vol-regime splits, 1-second lag tests, bootstrap CIs): thresholds logged before each run, failures kept as valid results, sealed 46-day holdout with counted access; a blind out-of-sample month then passed **7/7 pre-registered predictions** (per-symbol rank persistence ≈ 0.95).
- Caught artefacts before they reached the desk: event-grid sampling inflated correlations 5–10 $\times$ vs. time-grid; naive t-statistics overstated significance $\sim 4\times$ until corrected for cross-sectional/temporal dependence (effective-N); point-rank selection showed a winner's curse out-of-sample and was replaced by band selection.
- **Cost-aware backtests** (per-leg fees, colocation latency, lag retention) settled the viable monetization channel; production feature set fixed by exhaustive 511-subset evaluation; wrote the **A/B pilot** spec for desk approval.
- **Reproducible and traceable:** per-symbol-day Parquet caches, resumable batch jobs, one-command scripts regenerating every reported table, pre-registration log, per-phase reports, desk presentation, full hand-off document.

Machine Learning Engineer Intern

June 2025 – Sep. 2025

Riskoptima Hedge Fund

Istanbul, Turkey

- Modeled **volatility skew** and Greeks for 0-DTE SPX options and vertical spreads; backtested the strategies.
- Built a **VAE** (3D latent space) to synthesize VXX series for black-swan stress tests of trading algorithms.

PROJECTS & RESEARCH

Low-Latency Market Data Processor | C++, Multithreading, Optimization

Jan. 2025 – Sep. 2025

- High-throughput C++ engine for simulated exchange tick data: **lock-free ring buffer** between ingestion and processing threads (no mutex spikes), cache-coherent data layout, measured speedups over STL containers.

LLM-Guided Symbolic Feature Search | Python, LLM-in-the-Loop, Evolutionary Search

Jul. 2026 – Present

- FunSearch-style loop, LLM as mutation operator over feature expressions: 65 candidates in two waves scored on **pre-registered criteria** (partial correlation vs. known features) over a sealed train/holdout split.
- 7/8 finalists replicated on holdout (retention 0.91–1.16); one was rejected as a training-set artefact (11% retention); added an automatic coverage gate after finding that constant-valued series silently corrupt rank correlations.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, SciPy, Scikit-learn), C++ (STL, memory management, multithreading), SQL

Validation & Testing: Pre-registered design, sealed holdout / blind OOS protocols, placebo and permutation tests, bootstrap CIs, effective-N inference, cost-aware backtesting, data-quality gates

Markets & Microstructure: Full-depth (ITCH) order-book data, order-book imbalance / order-flow features, tick-size regimes, execution costs, HFT and low-latency concepts, options pricing (Black-Scholes, 0-DTE), volatility surface

Data, ML & Tooling: Multi-TB Parquet pipelines, resumable batch jobs with per-unit caching, time-series analysis, VAE, RL, Monte Carlo / MCMC, Git, Docker, Linux, CMake, GDB, LaTeX

ADDITIONAL INFORMATION

Languages: Turkish (Native), English (IELTS 7.0 – Professional)

Awards & Interests: ICYSS-2019 International Conference Presentation Award; Aikido 3rd Kyu (Hombu Dojo, Japan)

Certifications: SPL Level 2 – In Preparation